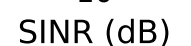


CRN Power Control: TD3 vs DDPG vs CAMO-TD3 Nakagami-m Fading Channel Performance Report

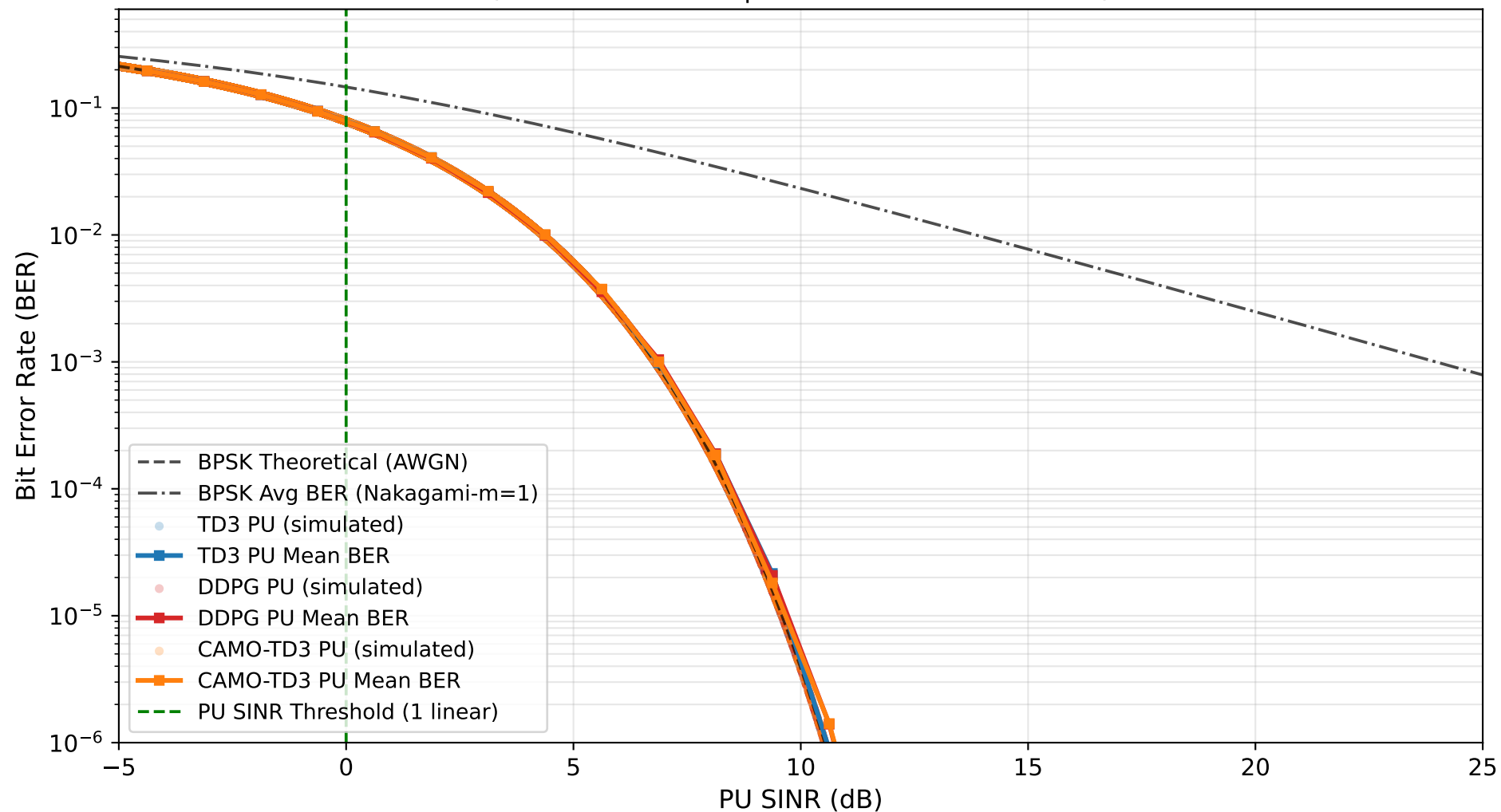
Nakagami-m = 1.0 | Episodes = 1300 | Steps/Episode = 200 | SINR Threshold = 1.0 dB

Metric	TD3	DDPG	CAMO-TD3	Winner
Avg Reward (last 100 ep)	306.9104	257.8817	312.9849	CAMO-TD3
SU Throughput (bits/s/Hz)	2.3195	1.9437	2.3771	CAMO-TD3
PU Throughput (bits/s/Hz)	0.8236	1.1484	0.7936	DDPG
Outage Probability	0.2601	0.3609	0.2561	CAMO-TD3
Average BER	0.061300	0.089923	0.059463	CAMO-TD3
Training Time (s)	5875.8	5492.4	13107.4	-

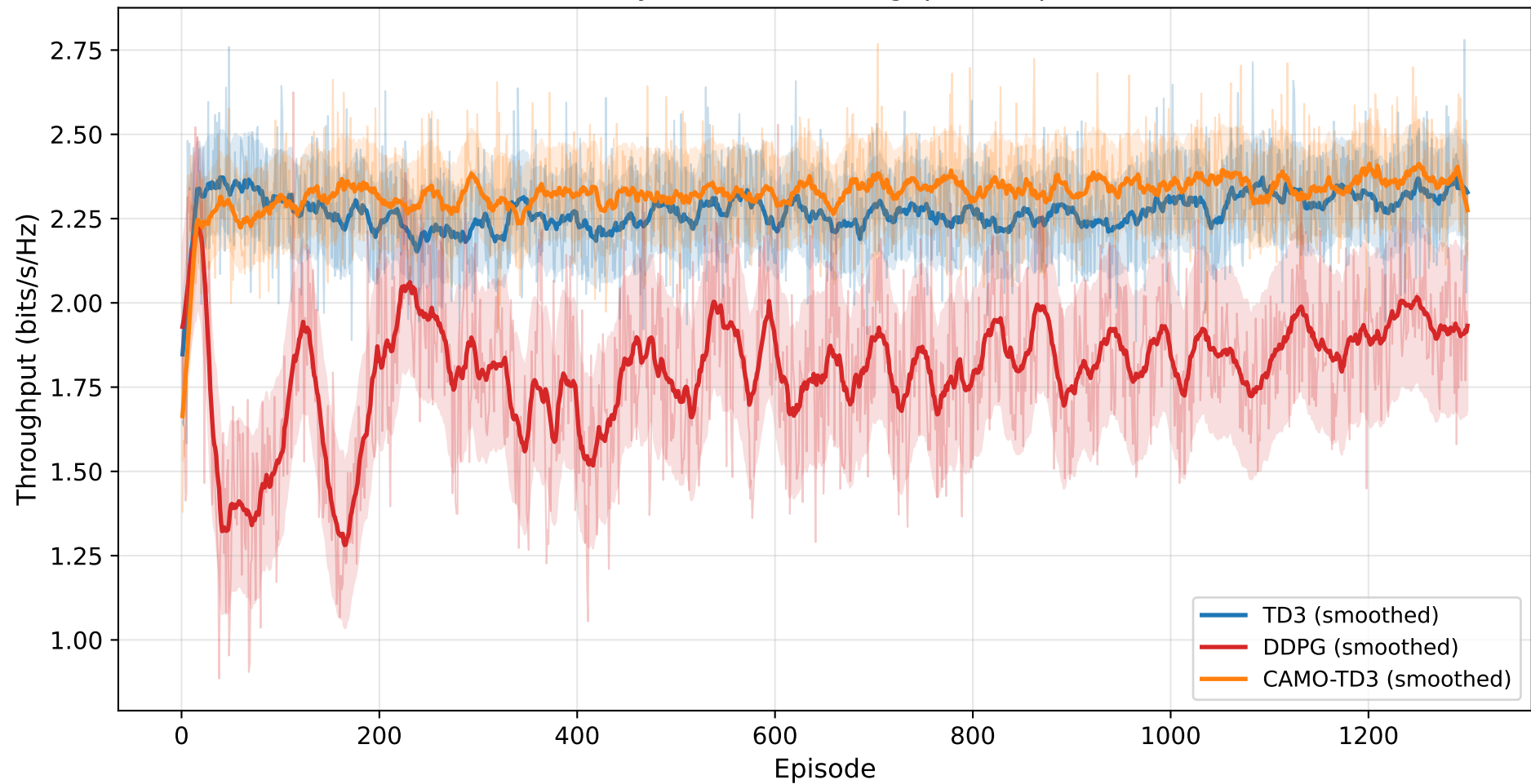
(BPSK Modulation, Secondary User)



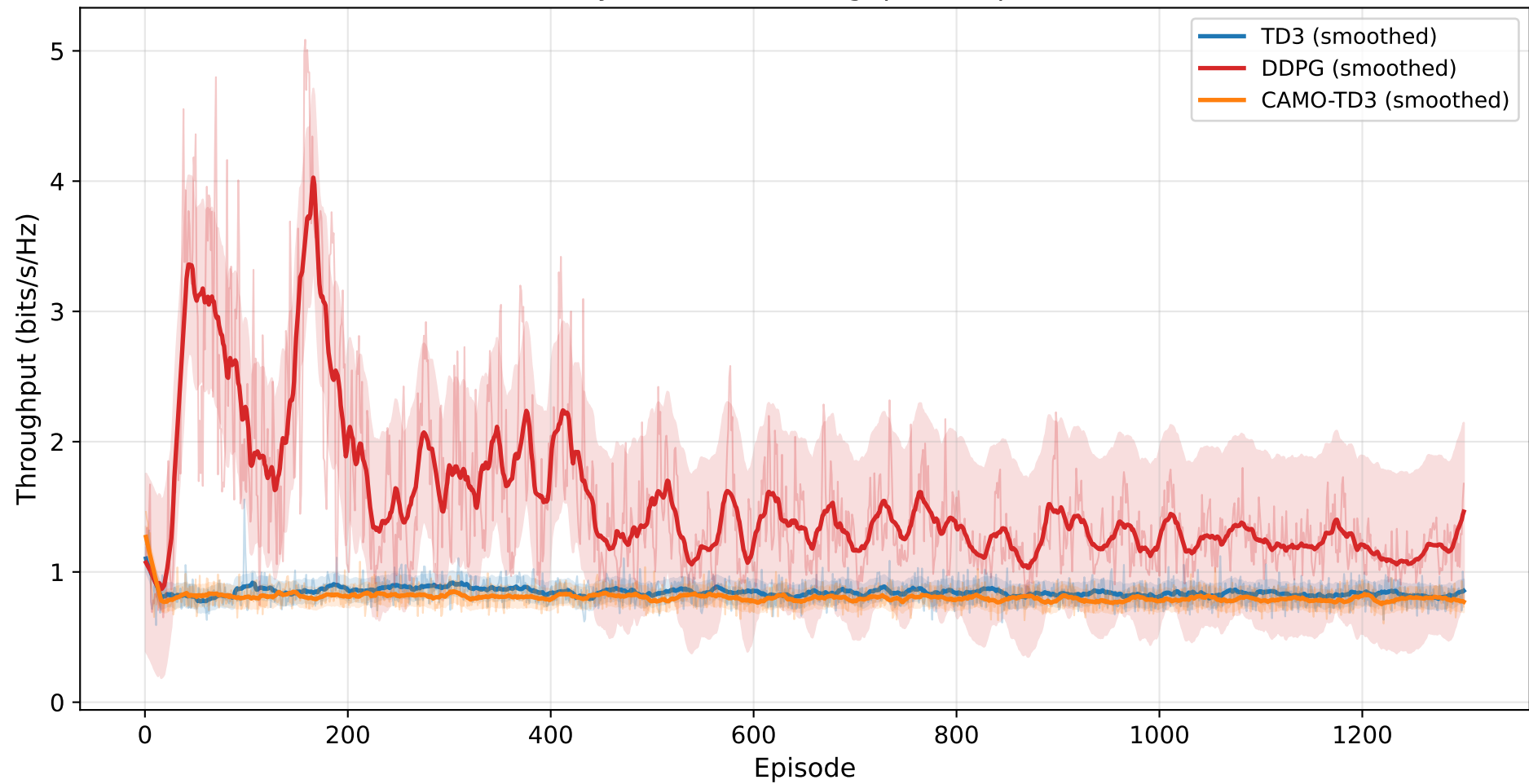
Primary User SINR vs BER - Nakagami-m=1 Fading Channel
(BPSK Modulation | Effect of SU Interference)



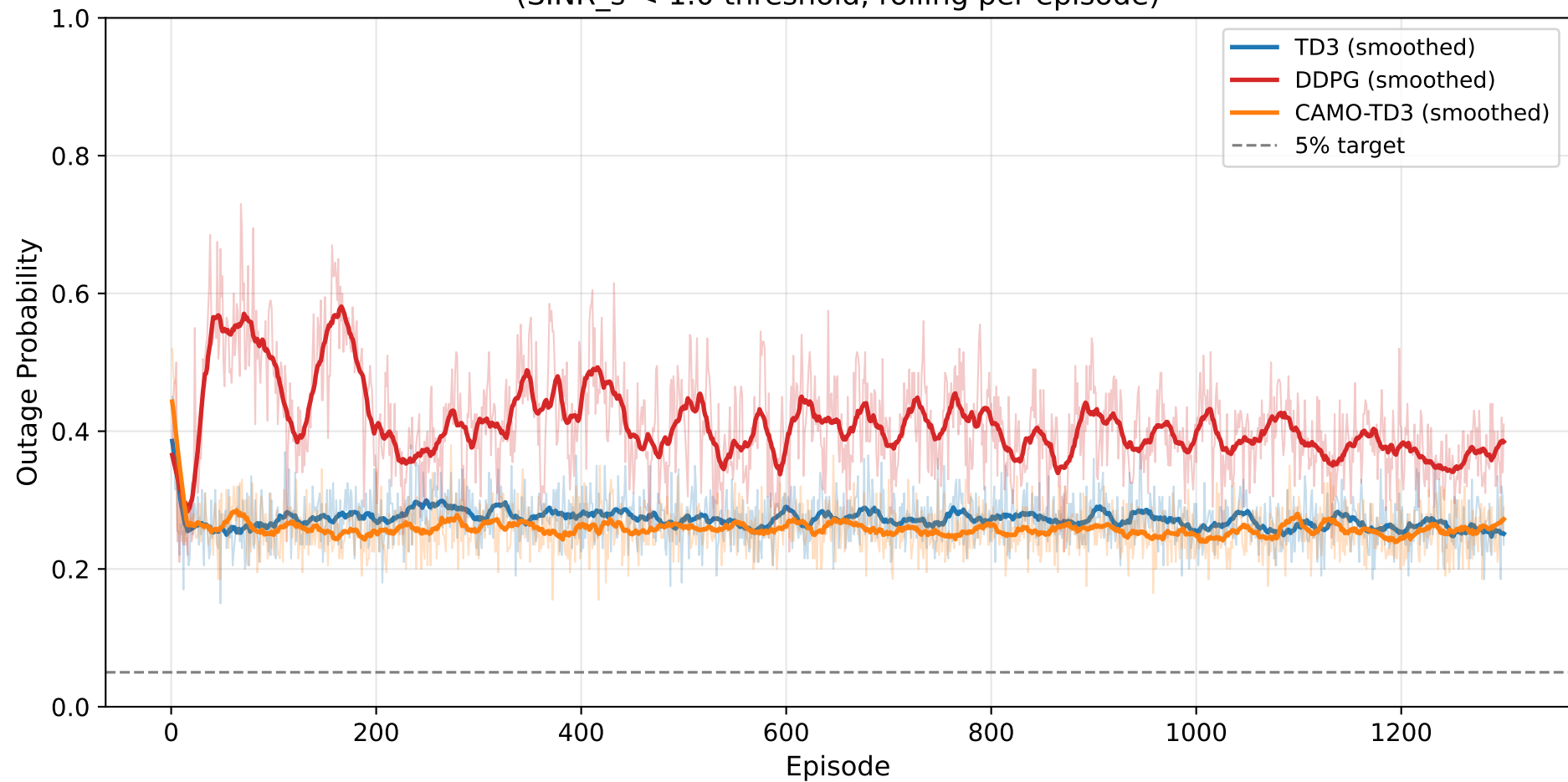
Secondary User (SU) Throughput vs Episodes



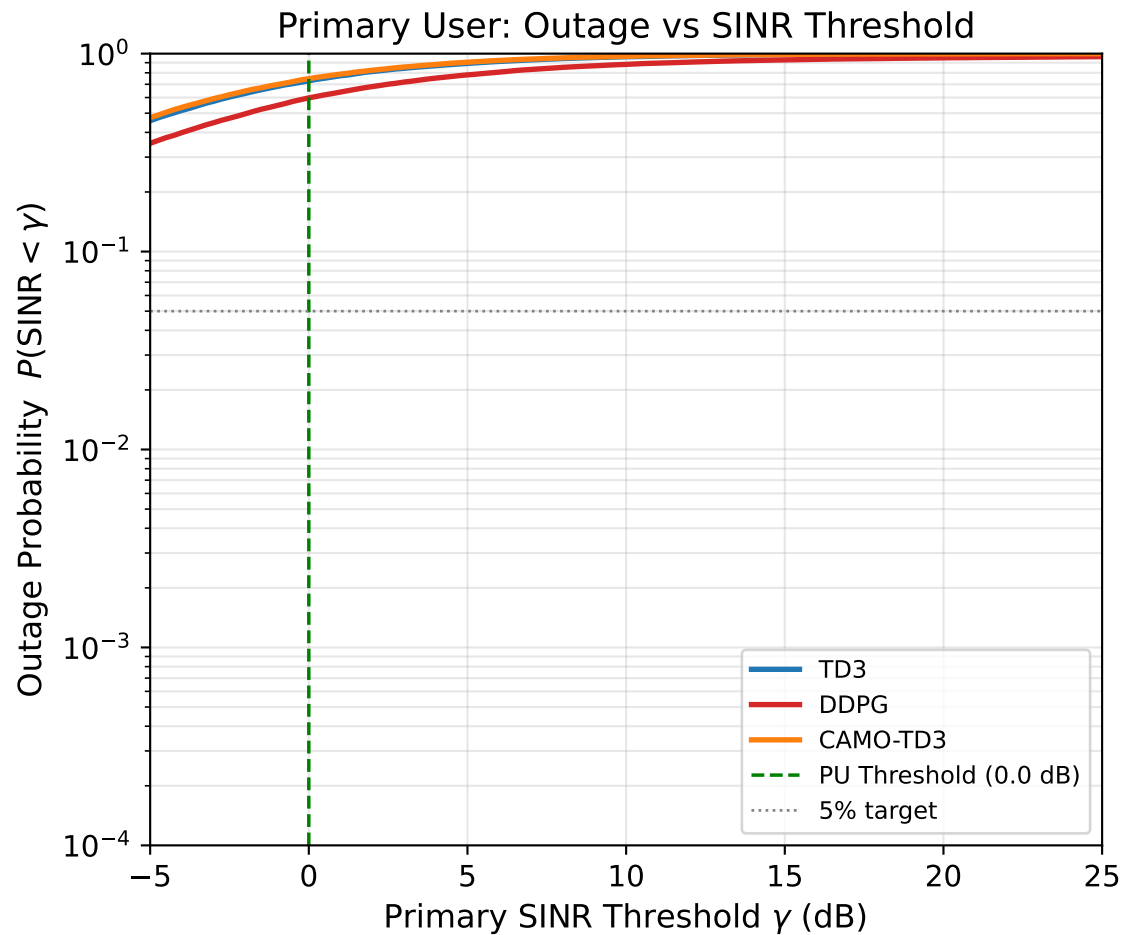
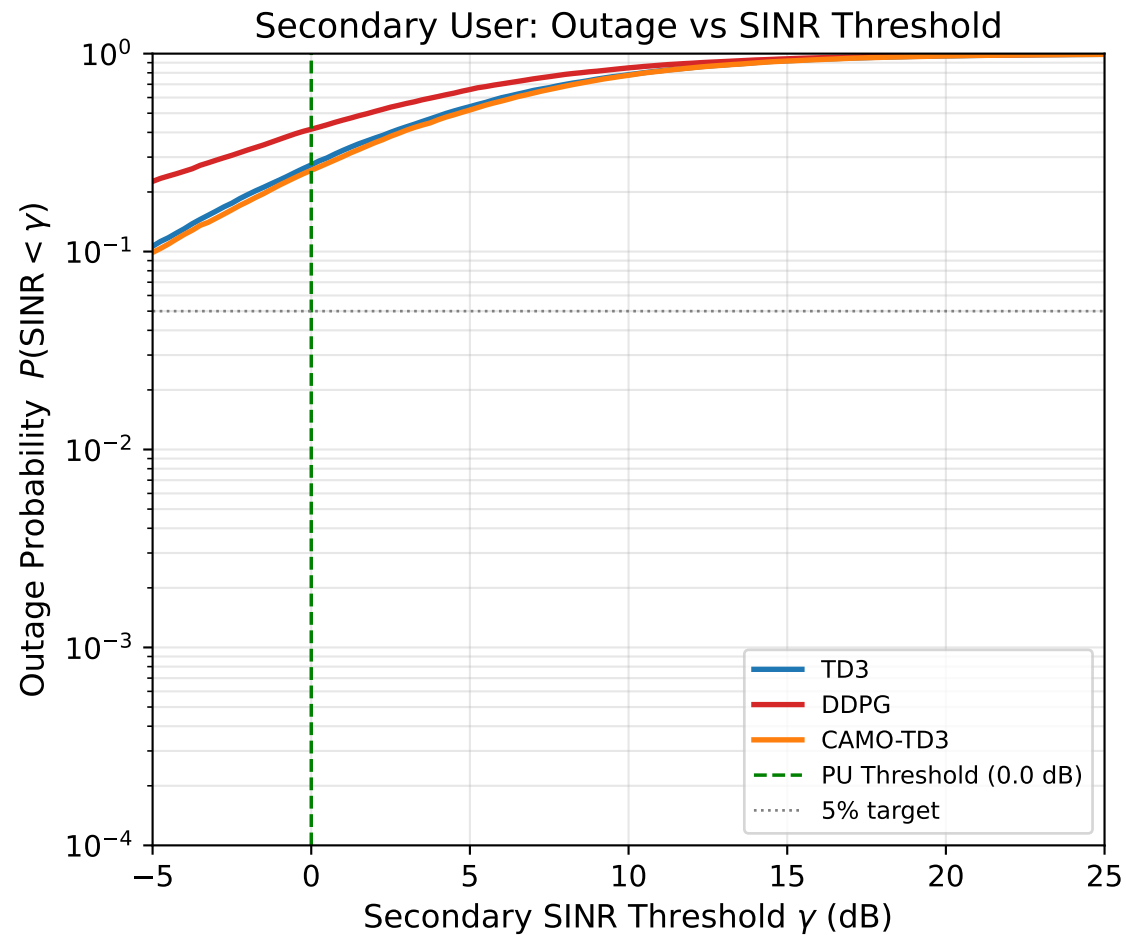
Primary User (PU) Throughput vs Episodes



Outage Probability vs Episodes
(SINR_s < 1.0 threshold, rolling per episode)

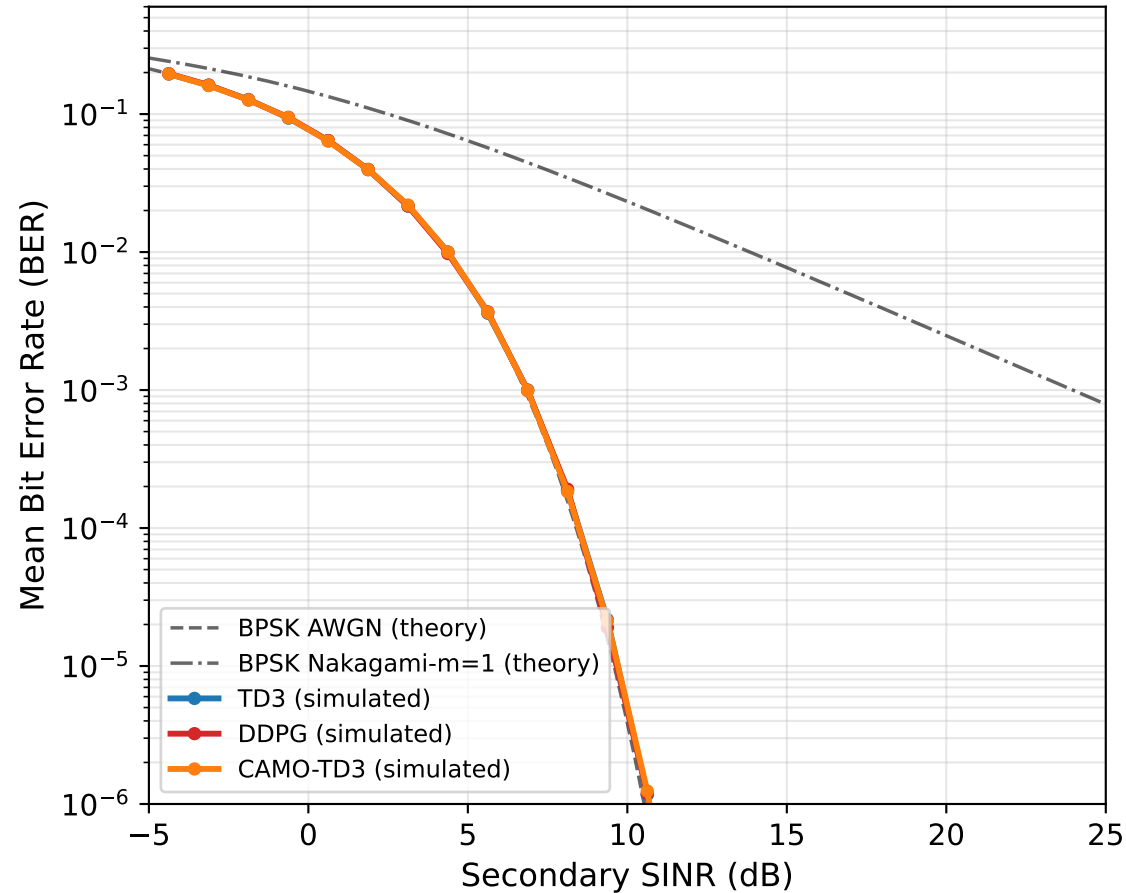


Outage Probability vs SINR Threshold (Empirical CDF) — Imperfect CSI

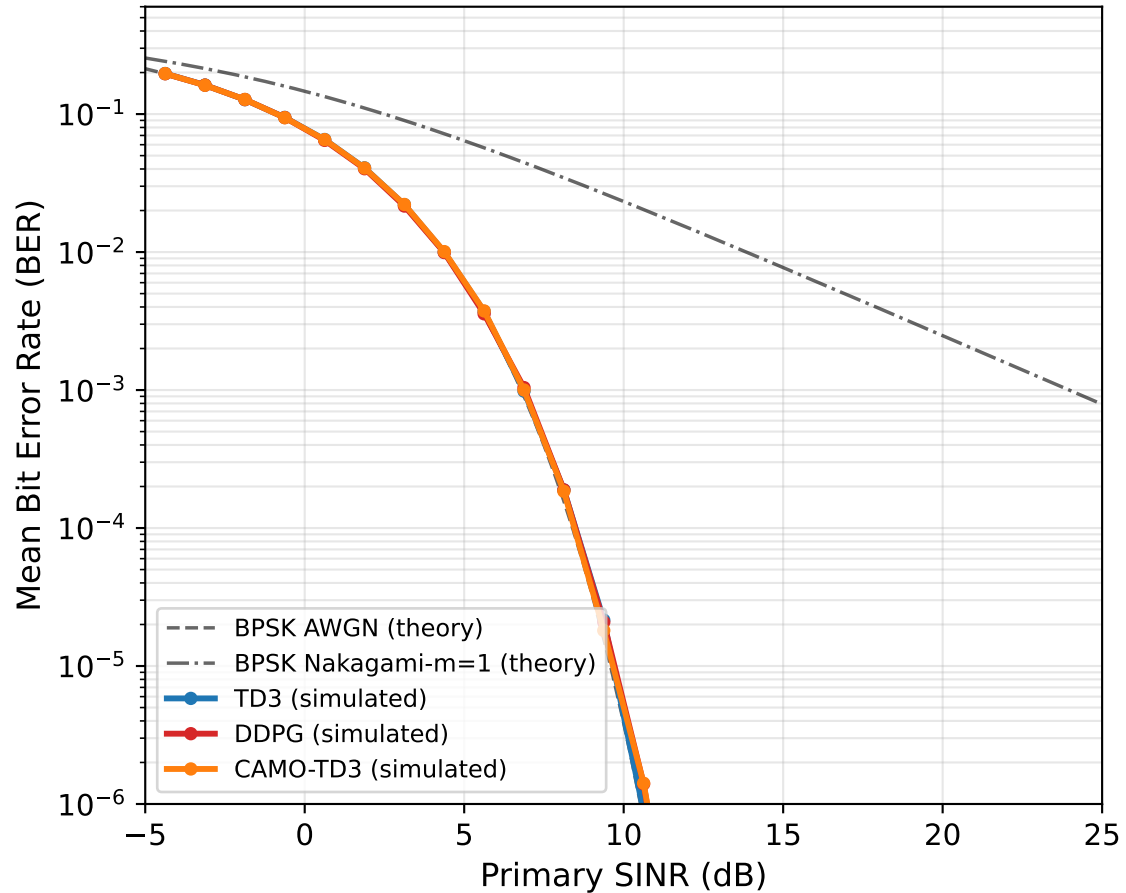


BER vs SINR — Algorithm Comparison under Imperfect CSI

Secondary User: BER vs SINR

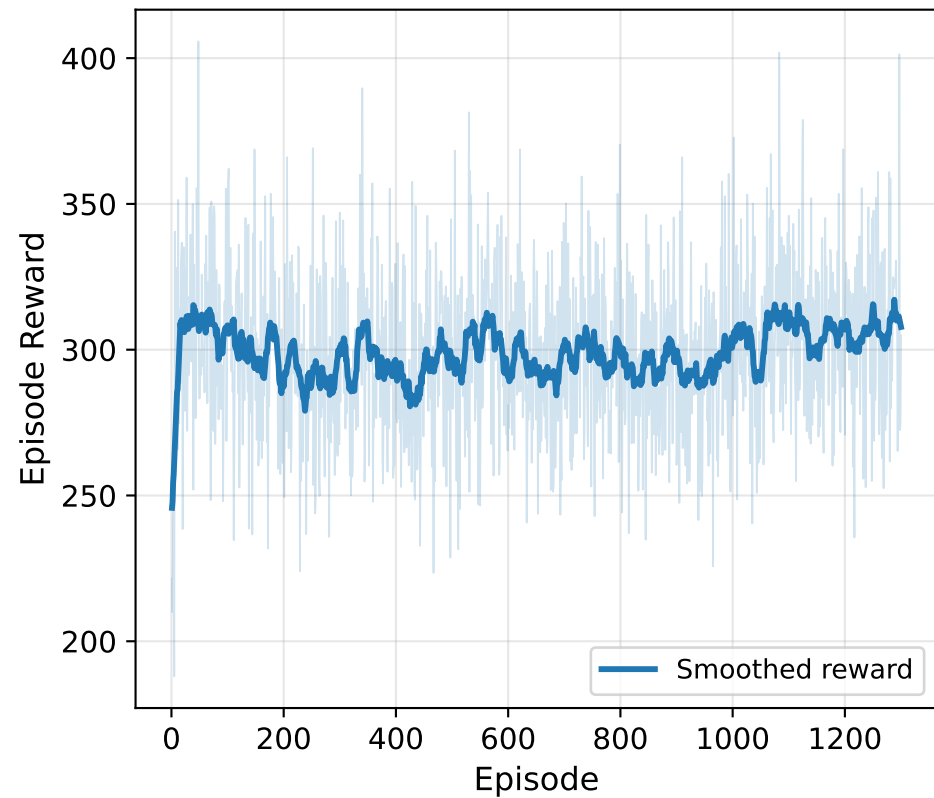


Primary User: BER vs SINR

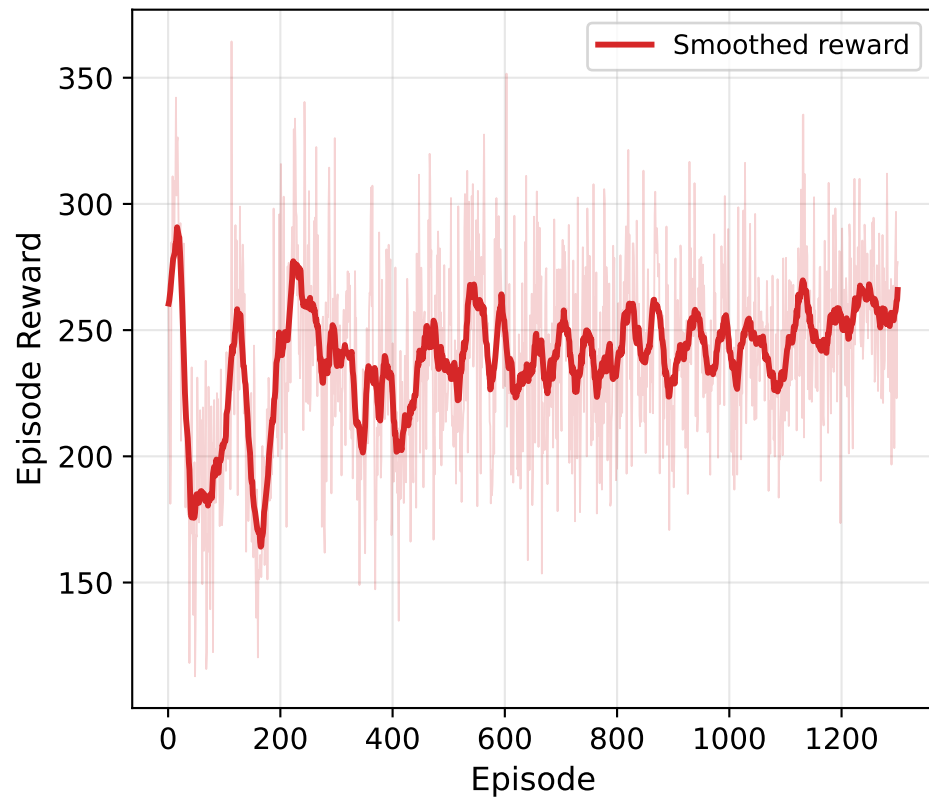


Episode Reward Curves - TD3 vs DDPG vs CAMO-TD3

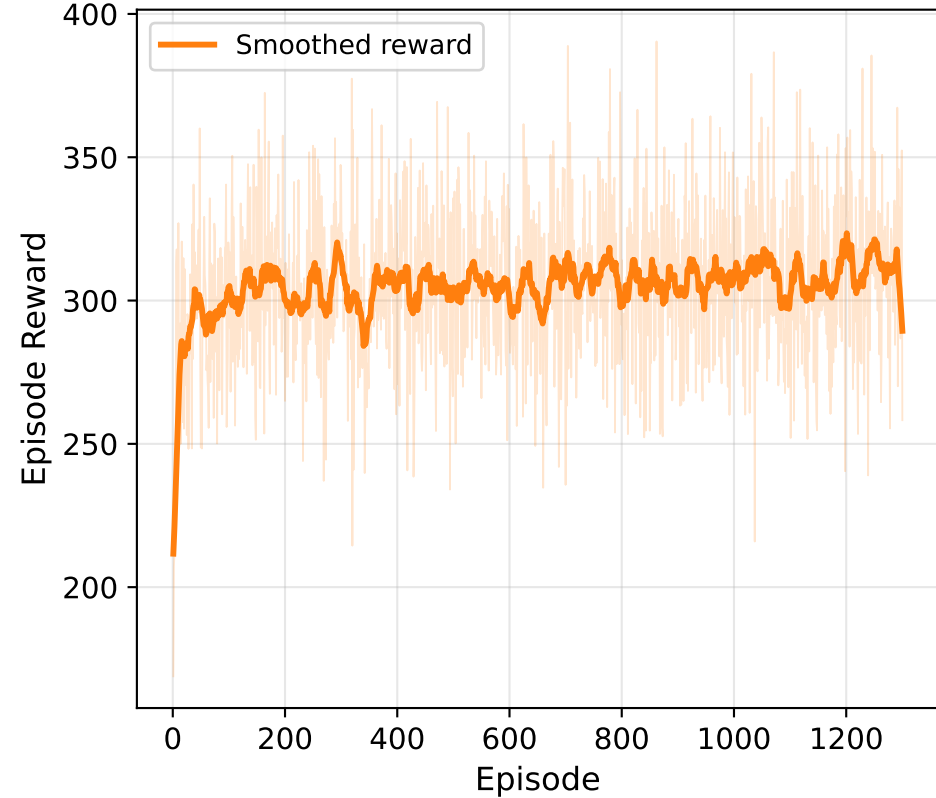
TD3 Reward Curve



DDPG Reward Curve



CAMO-TD3 Reward Curve



TD3 vs DDPG vs CAMO-TD3 - Reward Comparison

